

Part A: Order of Operations*Evaluate each expression completely.*

1. $(-3 + 16)^2 \div 1$

9. $(9 + 12)^2 \div 7$

2. $6^2 - 2 \cdot 9$

10. $-15 - 4 \cdot ((-6) + (-15))$

3. $(7 + 10)^2 \div 1$

11. $-5 - 11 \cdot (9 + (-5))$

4. $3 - 3 \cdot (7 + 3)$

12. $7^2 - 16 \cdot 10$

5. $3^2 - (-18) \cdot 6$

13. $(-5)^2 - 7 \cdot 5$

6. $7^2 - 10 \cdot 11$

14. $-16 - 9 \cdot ((-6) + (-16))$

7. $-11 - 12 \cdot (9 + (-11))$

15. $8 - 12 \cdot ((-7) + 8)$

8. $15 - 5 \cdot ((-4) + 15)$

16. $-2 - 8 \cdot (7 + (-2))$

Part B: Fractions*Evaluate. Write every answer in lowest terms.*

1. $\frac{5}{6} + \frac{2}{3}$

4. $\frac{1}{4} + \frac{1}{5}$

2. $\frac{1}{2} + 1\frac{2}{3}$

5. $1\frac{2}{9} \cdot \frac{7}{8}$

3. $2 + \frac{2}{3}$

6. $1\frac{5}{6} \div \frac{5}{12}$

10. $2\frac{1}{4} + 6$

7. $1\frac{3}{4} \div \frac{7}{9}$

11. $4\frac{4}{5} - 4\frac{1}{6}$

8. $\frac{1}{2} \div 1\frac{1}{4}$

12. $5 - 2\frac{5}{6}$

9. $1\frac{7}{8} + 5\frac{1}{3}$

Part C: Proportions

Solve each proportion for x .

1. $\frac{7}{14} = \frac{14}{x}$

5. $\frac{9}{14} = \frac{36}{x}$

2. $\frac{5}{13} = \frac{x}{78}$

6. $\frac{2}{x} = \frac{4}{28}$

3. $\frac{9}{4} = \frac{x}{8}$

7. $\frac{x}{7} = \frac{80}{56}$

4. $\frac{10}{15} = \frac{40}{x}$

8. $\frac{10}{10} = \frac{x}{60}$

Part D: Unit Rates

Find the unit rate, or determine the better price per unit.

1. Mia pays \$126 for 7 granola bars. Find the price per granola bar.

2. Kai pays \$140 for 10 erasers. Find the price per eraser.

3. Kai pays \$156 for 12 bagels. Find the price per bagel.
4. Omar pays \$63 for 9 juice boxes. Find the price per juice box.
5. Option A: \$50 for 10 bagels. Option B: \$120 for 6 bagels. Which option is the better price per bagel?
6. Option A: \$204 for 12 notebooks. Option B: \$190 for 10 notebooks. Which option is the better price per notebook?
7. Option A: \$144 for 9 pencils. Option B: \$33 for 3 pencils. Which option is the better price per pencil?
8. Option A: \$16 for 4 bagels. Option B: \$128 for 8 bagels. Which option is the better price per bagel?

Part E: Classifying Numbers

Classify each number using all that apply - rational, irrational, whole, integer.

- | | |
|------------------|------------------|
| 1. -15 | 4. -16 |
| 2. 39 | 5. 40 |
| 3. $\frac{6}{7}$ | 6. $\frac{3}{7}$ |

7. -31

10. 37

8. $\sqrt{2}$

11. 34

9. $-3\frac{3}{4}$

12. -29

Part F: Exponent Rules

Simplify. Write answers with positive exponents.

1. $8x^5 \cdot 9x^5$

7. $(6x^9)^3$

2. $8x^5 \cdot 7x^3$

8. $(8x^3)^2$

3. $2x^4 \cdot 3x^2$

9. 6^{-1}

4. $5x^4 \cdot 5x^3$

10. $(6^1)^{-1}$

5. $(2x^7)^4$

11. $4 \cdot (6^4)^{-4}$

6. $(4x^9)^2$

12. $\frac{8}{3^{-1}}$

Part G: Roots

Evaluate or simplify each root.

1. $\sqrt{49}$

4. $\sqrt{121}$

2. $\sqrt{16}$

5. $\sqrt[3]{27}$

3. $\sqrt{64}$

6. $\sqrt[3]{8}$

7. $\sqrt[3]{125}$

10. $\sqrt{18}$

8. $\sqrt[3]{64}$

11. $\sqrt{45}$

9. $\sqrt{224}$

12. $\sqrt{96}$

Part H: Percent Proportion

Solve the percent proportion, or find each value.

1. $\frac{39/2}{65} = \frac{x}{100}$

5. 12% of 16

2. $\frac{288/25}{144} = \frac{x}{100}$

6. 30% of 128

3. $\frac{x}{135} = \frac{25}{100}$

7. 12% of 144

8. 5% of 400

4. $\frac{6}{x} = \frac{12}{100}$

Part I: Estimating Percents

Estimate using the friendly benchmark given.

1. Estimate 14% of 28 by rounding to 10% of 30.

2. Estimate 12% of 296 by rounding to 10% of 300.

3. Estimate 79% of 216 by rounding to 75% of 220.

4. Estimate 24% of 69 by rounding to 25% of 70.

Part J: Percent Change

Find the percent change from the first value to the second. An increase is positive, a decrease is negative.

1. from 50 to 56
2. from 10 to 7
3. from 10 to 8
4. from 60 to 48

Part K: Percent Applications

Solve each percent application problem.

1. A store marks up a tablet that costs \$45 by 20%. Find the new price.
2. A jacket that normally costs \$100 is on sale for 5% off. Find the sale price.
3. A store marks up a lamp that costs \$125 by 8%. Find the new price.
4. A lamp that normally costs \$175 is on sale for 12% off. Find the sale price.
5. A recipe calls for 40 grams of flour. A student measures 50 grams. Find the percent error.
6. A recipe calls for 40 grams of flour. A student measures 42 grams. Find the percent error.
7. A recipe calls for 12 grams of flour. A student measures 9 grams. Find the percent error.

8. A recipe calls for 20 grams of flour. A student measures 23 grams. Find the percent error.
9. Ethan earns a 5% commission on sales of \$160. Find Ethan's commission.
10. Leo earns a 40% commission on sales of \$315. Find Leo's commission.
11. Sam earns a 40% commission on sales of \$175. Find Sam's commission.
12. Tara earns a 30% commission on sales of \$80. Find Tara's commission.
13. Wes's bill comes to \$132. Including a 25% tip, find the total amount Wes pays.
14. Leo's bill comes to \$60. Including a 30% tip, find the total amount Leo pays.
15. Maya's bill comes to \$70. Including a 60% tip, find the total amount Maya pays.
16. Diego's bill comes to \$350. Including a 8% tip, find the total amount Diego pays.

Part L: Linear Expressions*Simplify, combine, or factor each expression.*

1. $10x + 11 + 7x + 5$

9. $(4x + 2) - (12x + 4)$

2. $7x + 11 + 3x - 12$

10. $(8x + 3) - (x + 12)$

3. $-8x + 6 + 5x - 8$

11. $(-4x + 2) - (5x + 11)$

4. $x + 4 + 4x - 3$

12. $(2x + 2) - (-2x + 5)$

5. $9(3x + 5) + 12x + 3$

13. $5x - 30$

6. $12(12x + 3) + 9x + 8$

14. $45x - 9$

7. $-3(2x + 12) + 9x - 10$

15. $-30x + 25$

8. $5(-2x - 2) - 2x - 11$

16. $12x + 28$

Part M: Solving Equations*Solve each equation for x . Show your steps.*

1. $7x + 9 = -40$

3. $3x - 6 = -18$

2. $4x + 4 = -28$

4. $2x + 4 = -12$

5. $9x - 9 = -45$

15. $-x - 12 = -5x + 4$

6. $5x - 3 = 42$

16. $7x + 12 = -6x - 27$

7. $7(8x + 7) - 6 = 547$

17. $x + 7 = 11x - 33$

8. $5(2x + 8) - 8 = -28$

18. $4x - 12 = -8x + 60$

9. $9(3x + 1) + 11 = -169$

19. $\frac{1}{4}x + 12 = \frac{19}{2}$

10. $8(6x + 2) - 4 = 252$

20. $\frac{x + 14}{2} + 14 = 23$

11. $7(9x + 2) - 8 = -120$

21. $\frac{x}{4} + 1 = -\frac{7}{4}$

12. $3(6x + 6) + 1 = 1$

22. $\frac{x}{2} - 4 = -\frac{1}{2}$

13. $-12x + 3 = 5x - 167$

23. $\frac{x - 1}{3} - 12 = -\frac{43}{3}$

14. $3x - 3 = -6x - 57$

24. $\frac{x}{2} + 12 = \frac{21}{2}$

Part N: Solving Inequalities*Solve each inequality for x .*

1. $-6x + 5 > 23$

9. $4x + 10 > 5x + 1$

2. $8x + 5 > 45$

10. $7x - 2 > -12x + 93$

3. $-9x + 6 < -75$

11. $12x - 2 \geq 6x + 58$

4. $-6x + 7 < 37$

12. $11x - 11 < 8x - 44$

5. $-6x + 7 < -17$

13. $-9x + 6 > -12x + 33$

6. $6x + 4 \geq 58$

14. $5x + 9 < -8x - 134$

7. $8x - 1 < -33$

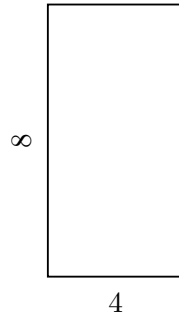
15. $10x - 7 \leq 11x - 9$

8. $6x - 8 \leq -56$

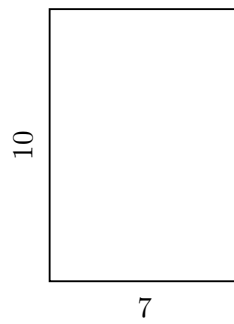
16. $-3x - 4 \leq 4x + 38$

Part O: Area*Find the area of each figure.*

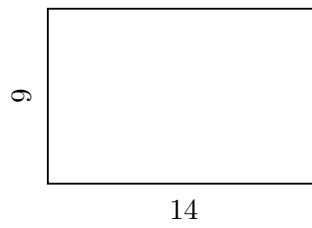
1. All measurements are in cm.



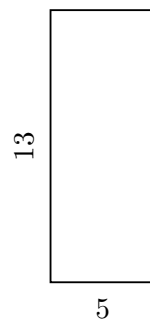
2. All measurements are in m.



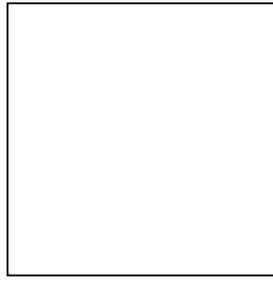
3. All measurements are in cm.



4. All measurements are in m.

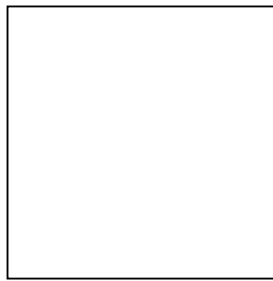


5. All measurements are in ft.



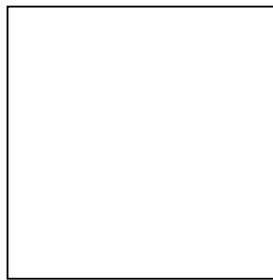
3

6. All measurements are in cm.



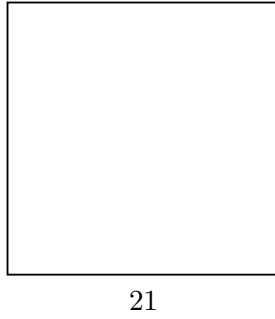
10

7. All measurements are in ft.

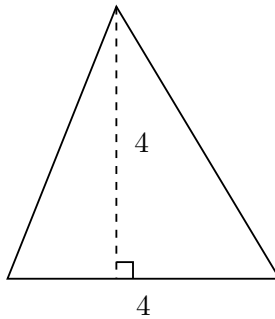


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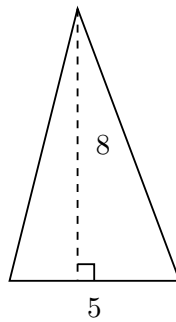
8. All measurements are in m.



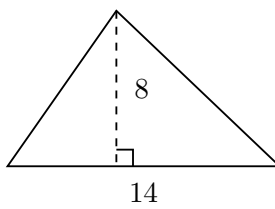
9. All measurements are in m.



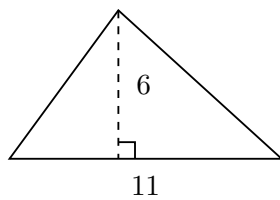
10. All measurements are in in.



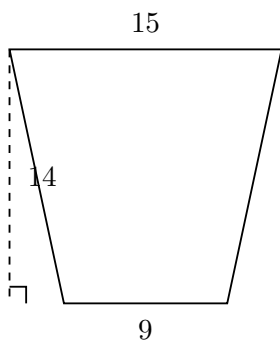
11. All measurements are in ft.



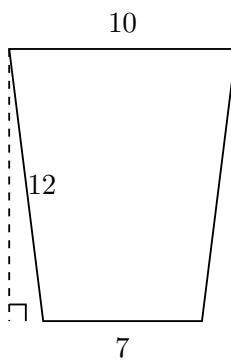
12. All measurements are in units.



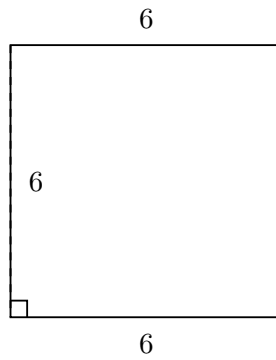
13. All measurements are in in.



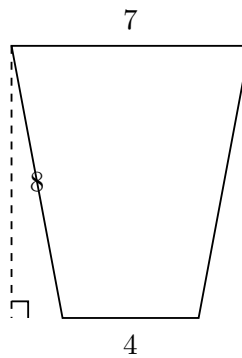
14. All measurements are in in.



15. All measurements are in ft.



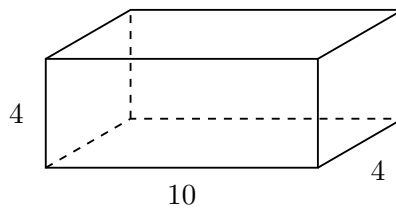
16. All measurements are in units.



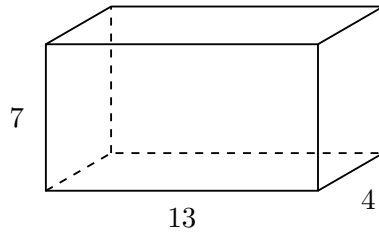
Part P: Volume of Prisms

Find the volume of each prism.

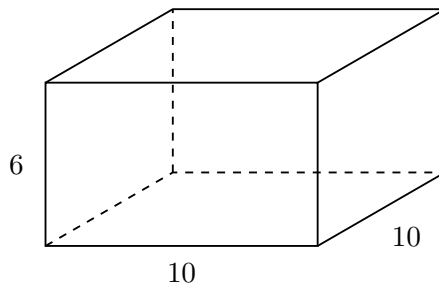
1. A rectangular prism. All measurements are in in.



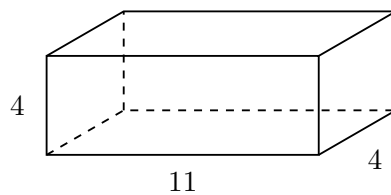
2. A rectangular prism. All measurements are in units.



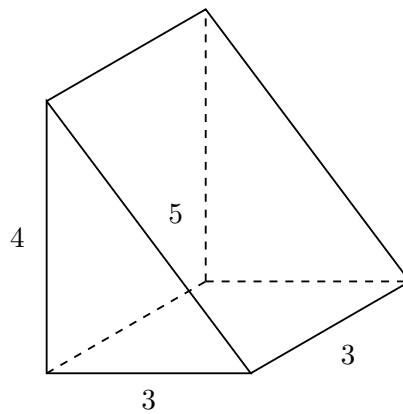
3. A rectangular prism. All measurements are in in.



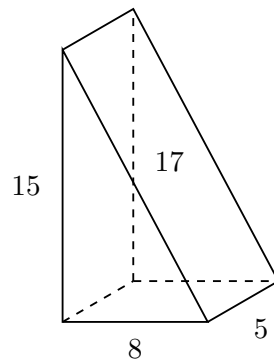
4. A rectangular prism. All measurements are in m.



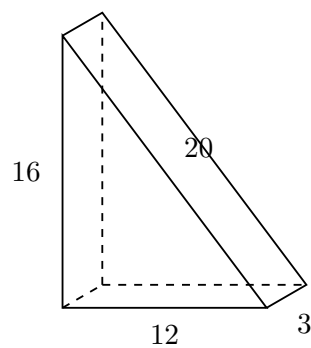
5. A triangular prism with a right-triangle base. All measurements are in cm.



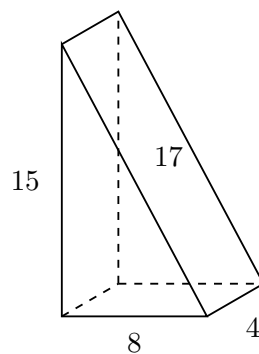
6. A triangular prism with a right-triangle base. All measurements are in m.



7. A triangular prism with a right-triangle base. All measurements are in in.



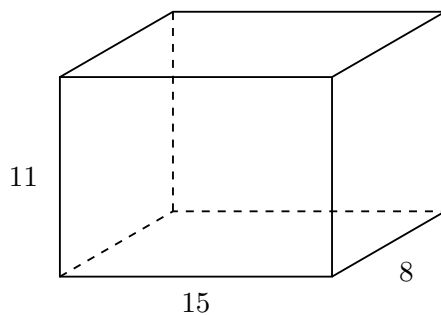
8. A triangular prism with a right-triangle base. All measurements are in units.



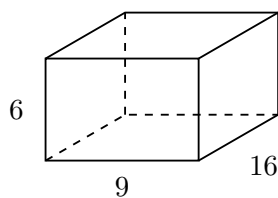
Part Q: Surface Area of Prisms

Find the surface area of each prism.

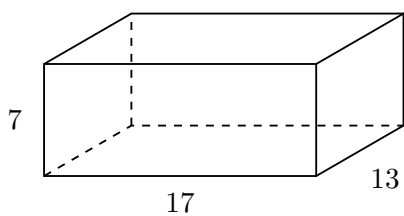
1. A rectangular prism. All measurements are in units.



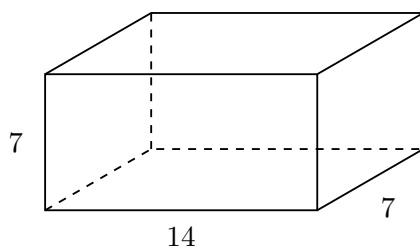
2. A rectangular prism. All measurements are in cm.



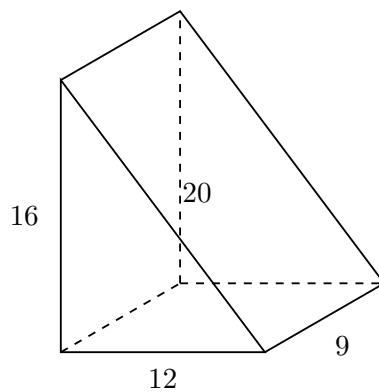
3. A rectangular prism. All measurements are in units.



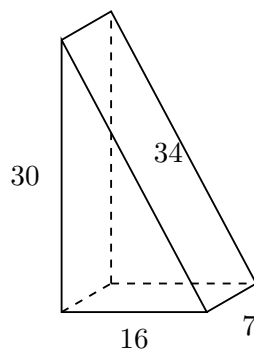
4. A rectangular prism. All measurements are in units.



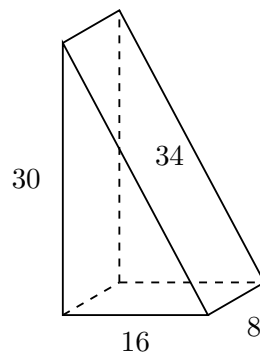
5. A triangular prism with a right-triangle base. All measurements are in cm.



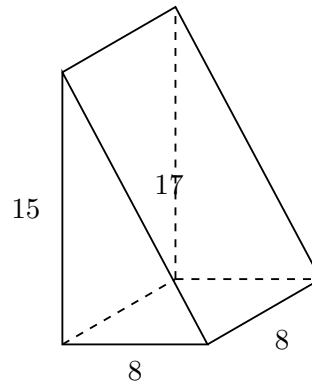
6. A triangular prism with a right-triangle base. All measurements are in units.



7. A triangular prism with a right-triangle base. All measurements are in units.



8. A triangular prism with a right-triangle base. All measurements are in m.



Answer Key

Part A: Order of Operations

- | | | |
|----------|------------|-----------|
| 1. 169 | 7. 13 | 13. -10 |
| 2. 18 | 8. -40 | 14. 182 |
| 3. 289 | 9. 63 | 15. -4 |
| 4. -27 | 10. 69 | 16. -42 |
| 5. 117 | 11. -49 | |
| 6. -61 | 12. -111 | |

Part B: Fractions

- | | | |
|-------------------|--------------------|---------------------|
| 1. $1\frac{1}{2}$ | 5. $1\frac{5}{72}$ | 9. $7\frac{5}{24}$ |
| 2. $2\frac{1}{6}$ | 6. $4\frac{2}{5}$ | 10. $8\frac{1}{4}$ |
| 3. $2\frac{2}{3}$ | 7. $2\frac{1}{4}$ | 11. $\frac{19}{30}$ |
| 4. $\frac{9}{20}$ | 8. $\frac{2}{5}$ | 12. $2\frac{1}{6}$ |

Part C: Proportions

- | | | |
|-------------|-------------|-------------|
| 1. $x = 28$ | 4. $x = 60$ | 7. $x = 10$ |
| 2. $x = 30$ | 5. $x = 56$ | 8. $x = 60$ |
| 3. $x = 18$ | 6. $x = 14$ | |

Part D: Unit Rates

- | | | |
|-------------------------|----------------------|-------------|
| 1. \$18 per granola bar | 4. \$7 per juice box | 7. Option B |
| 2. \$14 per eraser | 5. Option A | 8. Option A |
| 3. \$13 per bagel | 6. Option A | |

Part E: Classifying Numbers

- | | | |
|-----------------------------|-----------------------------|------------------------------|
| 1. Rational, Integer | 5. Rational, Integer, Whole | 9. Rational |
| 2. Rational, Integer, Whole | 6. Rational | 10. Rational, Integer, Whole |
| 3. Rational | 7. Rational, Integer | 11. Rational, Integer, Whole |
| 4. Rational, Integer | 8. Irrational | 12. Rational, Integer |

Part F: Exponent Rules

- | | | |
|---------------|----------------|------------------------------|
| 1. $72x^{10}$ | 5. $16x^{28}$ | 9. $\frac{1}{6}$ |
| 2. $56x^8$ | 6. $16x^{18}$ | 10. $\frac{1}{6}$ |
| 3. $6x^6$ | 7. $216x^{27}$ | 11. $\frac{1}{705277476864}$ |
| 4. $25x^7$ | 8. $64x^6$ | 12. 24 |

Part G: Roots

1. 7
2. 4
3. 8
4. 11

5. 3
6. 2
7. 5
8. 4

9. $4\sqrt{14}$
10. $3\sqrt{2}$
11. $3\sqrt{5}$
12. $4\sqrt{6}$

Part H: Percent Proportion

1. $x = 30$
2. $x = 8$
3. $x = \frac{135}{4}$

4. $x = 50$
5. $\frac{48}{25}$
6. $\frac{192}{5}$

7. $\frac{432}{25}$
8. 20

Part I: Estimating Percents

1. 3
2. 30

3. 165
4. $\frac{35}{2}$

Part J: Percent Change

1. 12%
2. -30%

3. -20%
4. -20%

Part K: Percent Applications

1. 54 dollars
2. 95 dollars
3. 135 dollars
4. 154 dollars
5. 25
6. 5

7. 25
8. 15
9. 8 dollars
10. 126 dollars
11. 70 dollars
12. 24 dollars

13. 165 dollars
14. 78 dollars
15. 112 dollars
16. 378 dollars

Part L: Linear Expressions

1. $17x + 16$
2. $10x - 1$
3. $-3x - 2$
4. $5x + 1$
5. $39x + 48$
6. $153x + 44$

7. $3x - 46$
8. $-12x - 21$
9. $-8x - 2$
10. $7x - 9$
11. $-9x - 9$
12. $4x - 3$

13. $5(x - 6)$
14. $9(5x - 1)$
15. $5(-6x + 5)$
16. $4(3x + 7)$

Part M: Solving Equations

1. $x = -7$
2. $x = -8$
3. $x = -4$
4. $x = -8$
5. $x = -4$
6. $x = 9$
7. $x = 9$
8. $x = -6$

9. $x = -7$
10. $x = 5$
11. $x = -2$
12. $x = -1$
13. $x = 10$
14. $x = -6$
15. $x = 4$
16. $x = -3$

17. $x = 4$
18. $x = 6$
19. $x = -10$
20. $x = 4$
21. $x = -11$
22. $x = 7$
23. $x = -6$
24. $x = -3$

Part N: Solving Inequalities

- | | | |
|---------------|-----------------|-----------------|
| 1. $x < -3$ | 7. $x < -4$ | 13. $x > 9$ |
| 2. $x > 5$ | 8. $x \leq -8$ | 14. $x < -11$ |
| 3. $x > 9$ | 9. $x < 9$ | 15. $x \geq 2$ |
| 4. $x > -5$ | 10. $x > 5$ | 16. $x \geq -6$ |
| 5. $x > 4$ | 11. $x \geq 10$ | |
| 6. $x \geq 9$ | 12. $x < -11$ | |

Part O: Area

- | | | |
|------------------|---------------------|---------------------|
| 1. 32 square cm | 7. 4 square ft | 13. 168 square in |
| 2. 70 square m | 8. 441 square m | 14. 102 square in |
| 3. 126 square cm | 9. 8 square m | 15. 36 square ft |
| 4. 65 square m | 10. 20 square in | 16. 44 square units |
| 5. 9 square ft | 11. 56 square ft | |
| 6. 100 square cm | 12. 33 square units | |

Part P: Volume of Prisms

- | | | |
|--------------------|----------------|--------------------|
| 1. 160 cubic in | 4. 176 cubic m | 7. 288 cubic in |
| 2. 364 cubic units | 5. 18 cubic cm | 8. 240 cubic units |
| 3. 600 cubic in | 6. 300 cubic m | |

Part Q: Surface Area of Prisms

- | | | |
|---------------------|----------------------|----------------------|
| 1. 746 square units | 4. 490 square units | 7. 1120 square units |
| 2. 588 square cm | 5. 624 square cm | 8. 440 square m |
| 3. 862 square units | 6. 1040 square units | |